Assignment Name - Practice Assignment - GCP

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Batch - 16A

Date - Oct 31st, 2026

GIT Repo - https://github.com/saleeminfodevops/herovired.git

https://github.com/saleeminfodevops/herovired/tree/gcp/assignments/gcp

Description

Deploy a MERN application using Kubernetes. Make sure the proper documentation along with screenshots of the steps are mentioned in your document and pushed into a GIT repository.

Link to the application:

https://github.com/UnpredictablePrashant/SampleMERNwithMicroservices

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Submission Instructions:

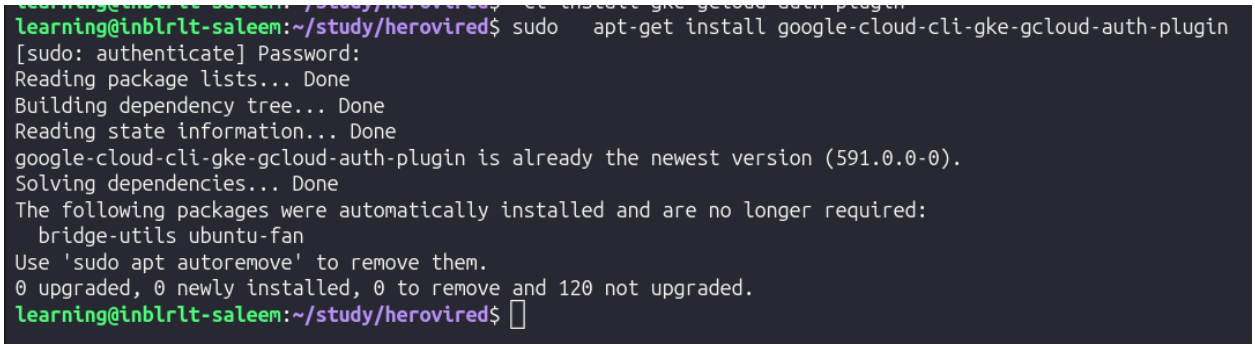
To submit your assignment, please follow these guidelines:

- Ensure that your assignment is fully completed.
- Push your code to a GitHub repository.
- Share the repository link by including it in a text, Word, or PDF file format.

Submit the file/text containing the repository link via Vlearn.

Step 1: Prerequisites & Clone the Repository

1. Install and initialize the Google Cloud CLI (`gcloud`).



Clone the repository

```
git clone
https://github.com/UnpredictablePrashant/SampleMERNwithMicroservices.git
cd SampleMERNwithMicroservices
```

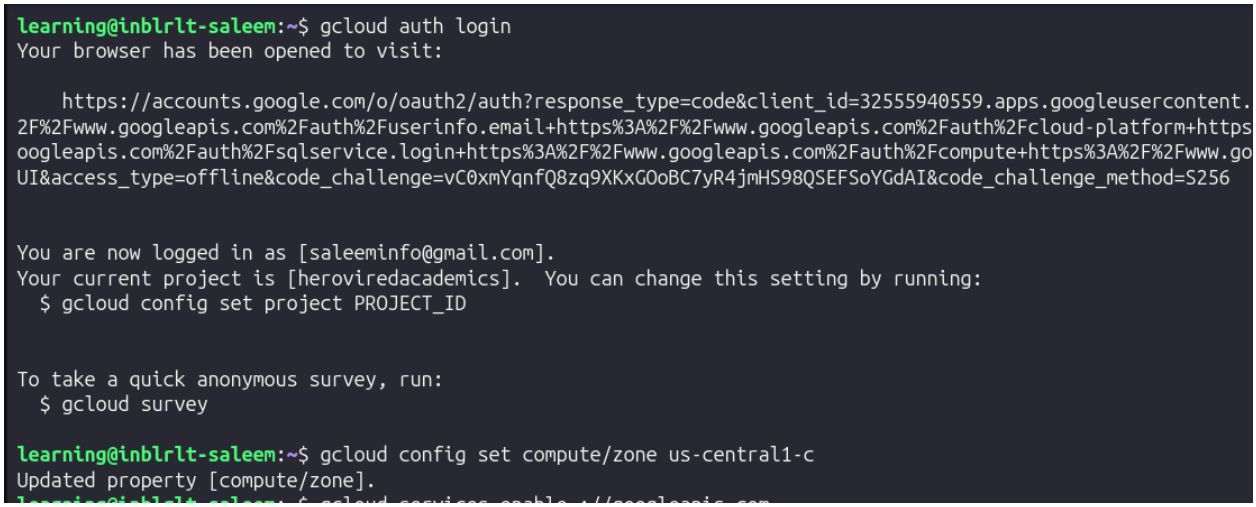
Step 2: Authenticate and Configure GCP Project

Log in to your GCP account via the terminal:

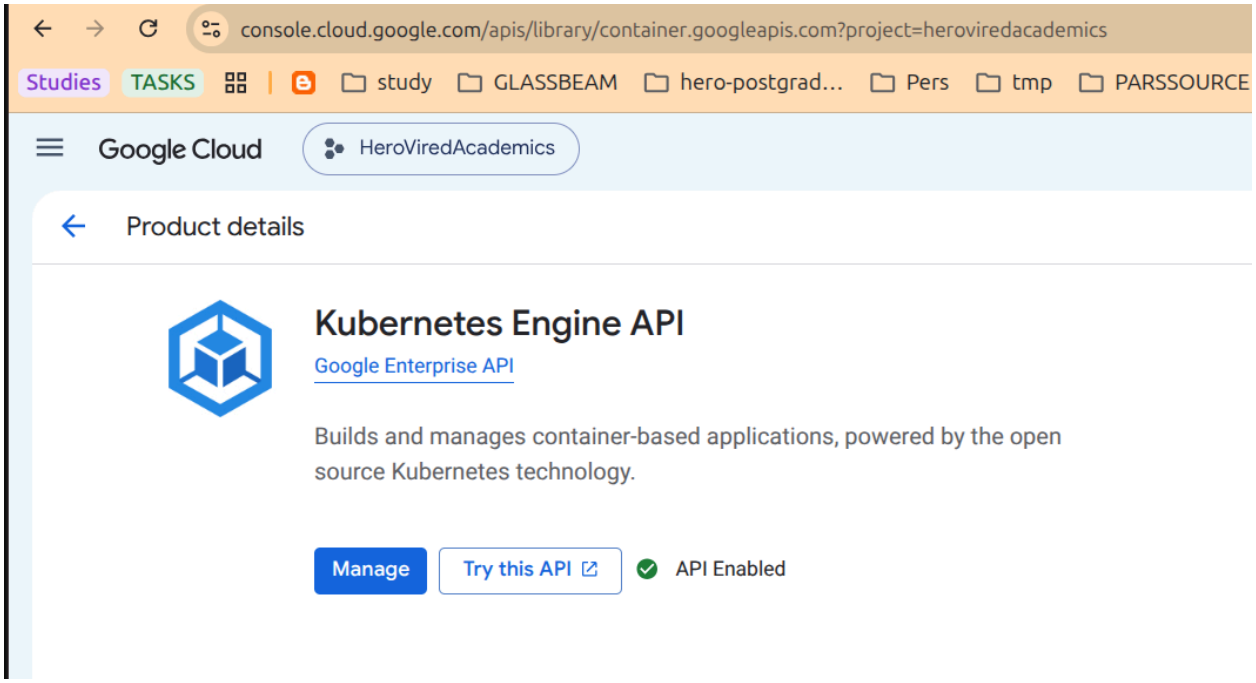
```
gcloud auth login
```

Set your target project ID and default compute region/zone:

```
gcloud config set project heroviredacademics
gcloud config set compute/zone us-central1-c
```



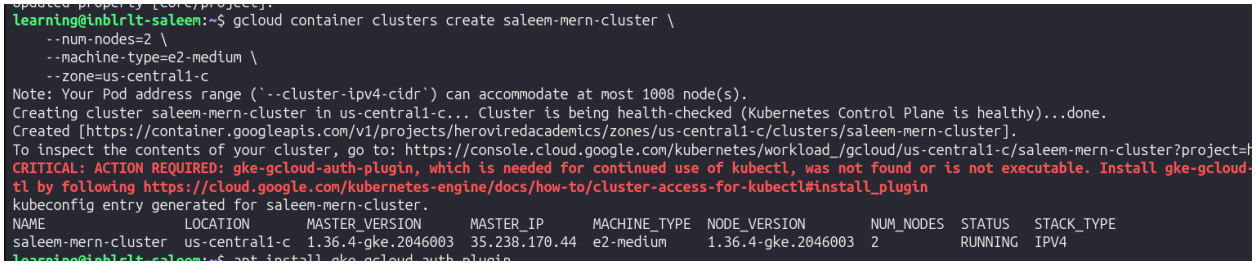
Enable the required Kubernetes Engine API (already enabled)



Step 3: Create a GKE Kubernetes Cluster **(The cluster is getting deleted)**

Create a standard GKE cluster using the `gcloud`

```
gcloud container clusters create saleem-mern-cluster \
    --num-nodes=2 \
    --machine-type=e2-medium \
    --zone=us-central1-c
```



```
learning@inblrlt-saleem:~$ sudo   apt-get install google-cloud-cli-gke-gcloud-auth-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following packages were automatically installed and are no longer required:
  bridge-utils ubuntu-fan
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  google-cloud-cli-gke-gcloud-auth-plugin
0 upgraded, 1 newly installed, 0 to remove and 120 not upgraded.
Need to get 4,149 kB of archives.
After this operation, 10.4 MB of additional disk space will be used.
Get:1 https://packages.cloud.google.com/apt cloud-sdk/main amd64 google-cloud-cli-gke-gcloud-auth
Fetched 4,149 kB in 2s (1,868 kB/s)
Selecting previously unselected package google-cloud-cli-gke-gcloud-auth-plugin.
(Reading database… 269562 files and directories currently installed.)
Preparing to unpack …/google-cloud-cli-gke-gcloud-auth-plugin_591.0.0-0_amd64.deb…
Unpacking google-cloud-cli-gke-gcloud-auth-plugin (591.0.0-0)…
Setting up google-cloud-cli-gke-gcloud-auth-plugin (591.0.0-0)…
```

Once the cluster is provisioned, get the authentication credentials so `kubectl` can communicate with your cluster

```
gcloud container clusters get-credentials mern-cluster --zone=us-central1-c
```

```
learning@inblrlt-saleem:~$ gcloud container clusters get-credentials saleem-mern-cluster --zone=us-central1-c
Fetching cluster endpoint and auth data.
kubeconfig entry generated for saleem-mern-cluster.
```

Verify the node status:

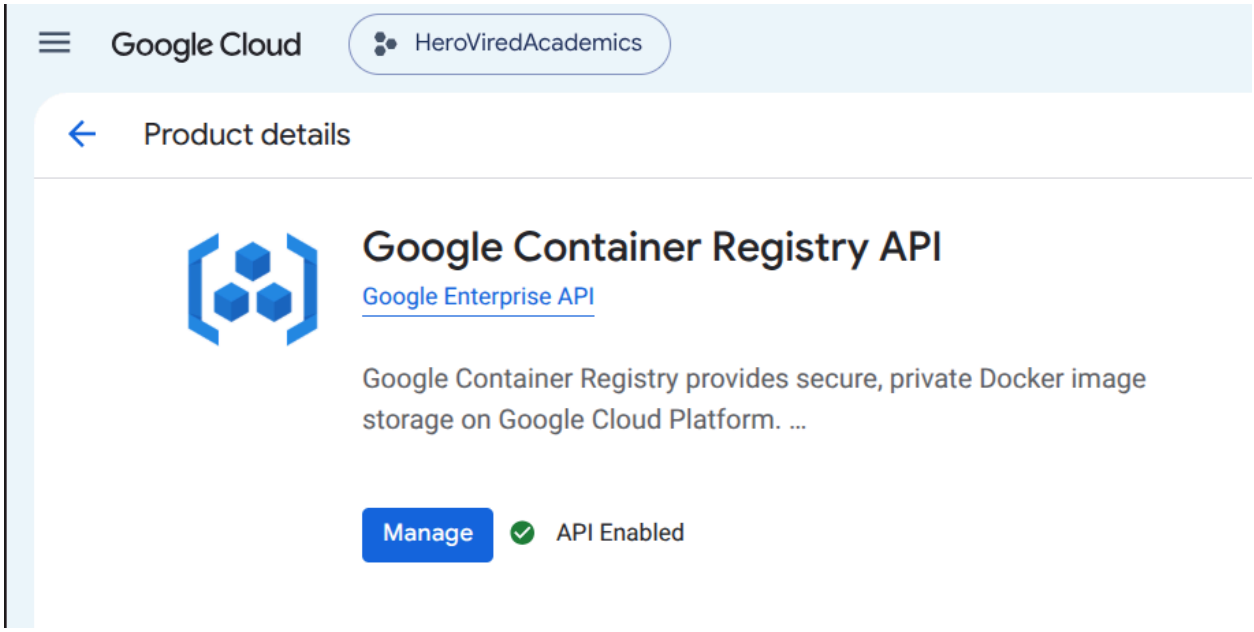
```
kubectl get nodes
```

```
learning@inblrlt-saleem:~$ kubectl get nodes
NAME                                                STATUS   ROLES    AGE    VERSION
gke-saleem-mern-cluster-default-pool-9a961138-dbpw   Ready    <none>   4m2s   v1.36.4-gke.2046003
gke-saleem-mern-cluster-default-pool-9a961138-krxr   Ready    <none>   4m3s   v1.36.4-gke.2046003
```

Step 4: Build and Push Docker Images to Artifact Registry (Already enabled)

Enable Artifact Registry in your project:

```
gcloud services enable ://googleapis.com
```



Create an Artifact Registry repository:

```
      gcloud artifacts repositories create saleem-repo
--repository-format=docker --location=us-central1
```

Build and tag your frontend and microservice backend Docker images. For example, configure the Docker to authenticate with GCP:

```
gourl="us-central1-docker.pkg.dev/heroviredacademics/saleem-repo"

docker build -t $gourl/saleem-mern-frontend:latest ./frontend
docker push $gourl/saleem-mern-frontend:latest
```

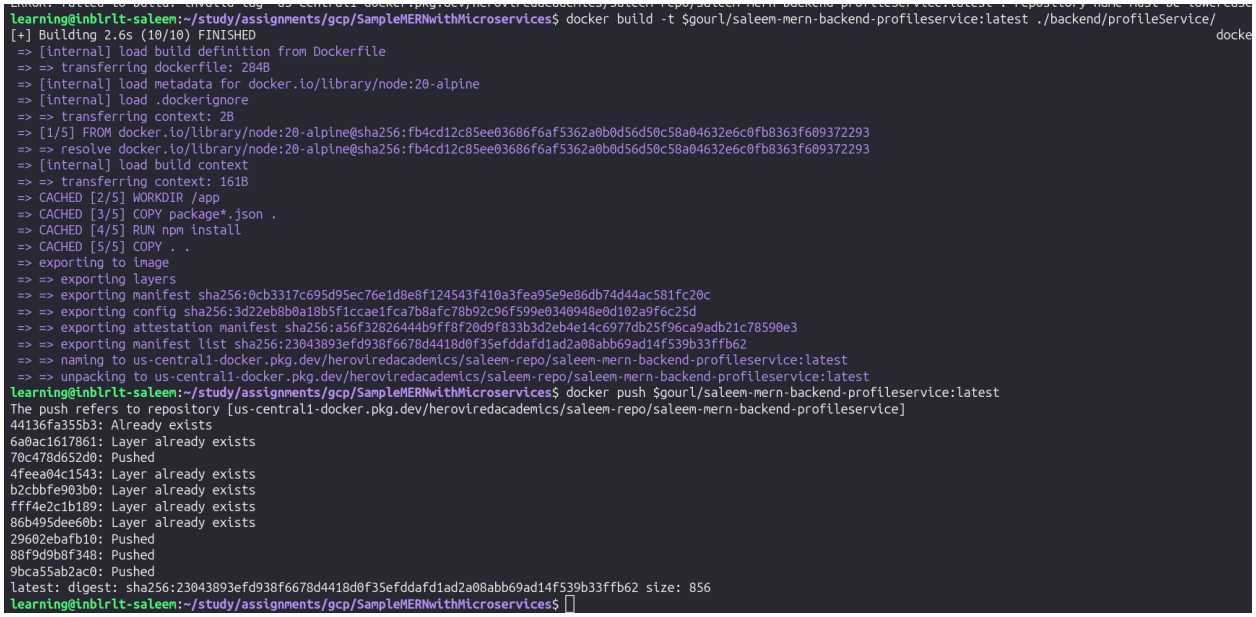
```
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$ gourl="us-central1-docker.pkg.dev/heroviredacademics/saleem-repo"
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$ docker build -t $gourl/saleem-mern-frontend:latest ./frontend
[+] Building 1.6s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 206B
 => [internal] load metadata for docker.io/library/node:20-alpine
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293
 => => resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293
 => [internal] load build context
 => => transferring context: 816B
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY package*.json ./
 => CACHED [4/5] RUN npm install
 => CACHED [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:6e074f30712918c7b1bd87cfc0a44b18f5ed7c964c44d9a1ce4ad969323ab03b
 => => exporting config sha256:81bd1b418b9a22dee3cf29f62c3c065785fd09df411d4d454de07bae43e6ff4b
 => => exporting attestation manifest sha256:7f337edef639e004b3a7d5c89eb0a84d0c7dcd6999fe7a109ff60c31b8f3bf05
 => => exporting manifest list sha256:70de8d6f013bc796eb68e2d86a275b2a2e3bab95602f056794e55167ea27d71f
 => => naming to us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-frontend:latest
 => => unpacking to us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-frontend:latest
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$ docker push $gourl/saleem-mern-frontend:latest
The push refers to repository [us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-frontend]
44136fa355b3: Pushed
6a0ac1617861: Pushed
4feea04c1543: Pushed
ffc46d013d25: Pushed
b2cbbfe903b0: Pushed
fff4e2c1b189: Pushed
86b495dee60b: Pushed
d966db3b2cfb: Pushed
a025ec224a60: Pushed
9d804559b738: Pushed
latest: digest: sha256:70de8d6f013bc796eb68e2d86a275b2a2e3bab95602f056794e55167ea27d71f size: 856
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$
```

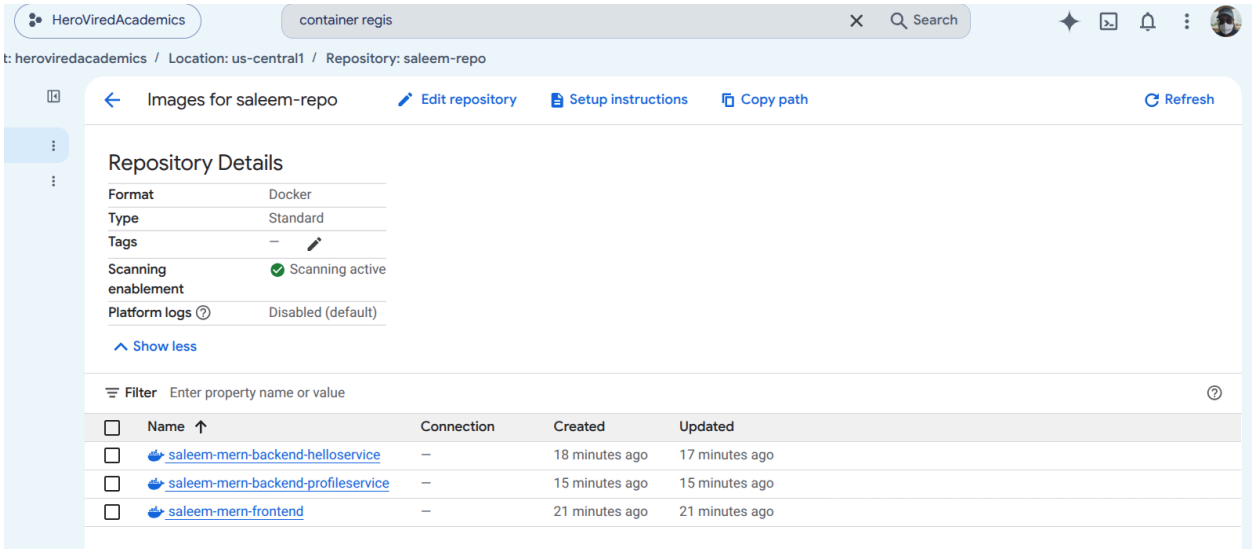
docker build -t $gourl/saleem-mern-backend-helloservice:latest ./backend/helloService/

docker push $gourl/saleem-mern-backend-helloservice:latest

```
Dockerfile  Jenkinsfile  index.js  package-lock.json  package.json
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$ docker build -t $gourl/saleem-mern-backend-helloservice:latest ./backend/helloService/
[+] Building 1.5s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 155B
 => [internal] load metadata for docker.io/library/node:20-alpine
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293
 => => resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293
 => [internal] load build context
 => => transferring context: 160B
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY package*.json .
 => CACHED [4/5] RUN npm install
 => CACHED [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:a6be2055a89cee0e38d86077e8e9e01f75cb54615d2ecac07f7f412fc4814313
 => => exporting config sha256:5c284c0c8c39f153d356b7206bd5eecf0931169d0fba82e509d20426ebe6e0ee
 => => exporting attestation manifest sha256:798ddfccc6365592fd029c72930e5e2811521603e44fa5f2c50ab71de061d8e7
 => => exporting manifest list sha256:48fd692bda8109466da059622170455a24e517d5d872d9d7e2c142c738c79125
 => => naming to us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-backend-helloservice:latest
 => => unpacking to us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-backend-helloservice:latest
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$ docker push $gourl/saleem-mern-backend-helloservice:latest
The push refers to repository [us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-backend-helloservice]
4feea04c1543: Layer already exists
44136fa355b3: Already exists
6a0ac1617861: Layer already exists
30a9eb19aaef: Pushed
b2cbbfe903b0: Layer already exists
fff4e2c1b189: Layer already exists
86b495dee60b: Layer already exists
14304f6bd869: Pushed
4950e7e33bc8: Pushed
d55bde13a491: Pushed
latest: digest: sha256:48fd692bda8109466da059622170455a24e517d5d872d9d7e2c142c738c79125 size: 856
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$
```

```
docker build -t $gourl/saleem-mern-backend-profileservice:latest
./backend/profileService/
docker push $gourl/saleem-mern-backend-profileservice:latest
```





Step 5: Apply Kubernetes Manifests

1. Inspect the Kubernetes deployment and service YAML manifests (`manifest.yml` or files inside the repository). Ensure the image paths inside the YAML point to your pushed GCR/Artifact Registry images.

Deploy the objects to your GKE cluster:

```
kubectl apply -f manifest.yml
```

Verify that your deployments, pods, and services are running properly:

```
kubectl get deployments
kubectl get pods
kubectl get svc
```

Step 6: Access Your MERN Application

Check the external IP address or LoadBalancer endpoint assigned to your frontend service:

```
kubectl get svc
```

- Open the external IP or LoadBalancer URL in your browser to interact with the MERN application.